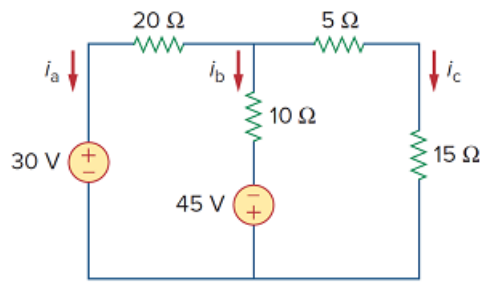


Problem

Use mesh analysis to obtain i_a , i_b , and i_c in the circuit in Fig. 3.84.

Figure 3.84:

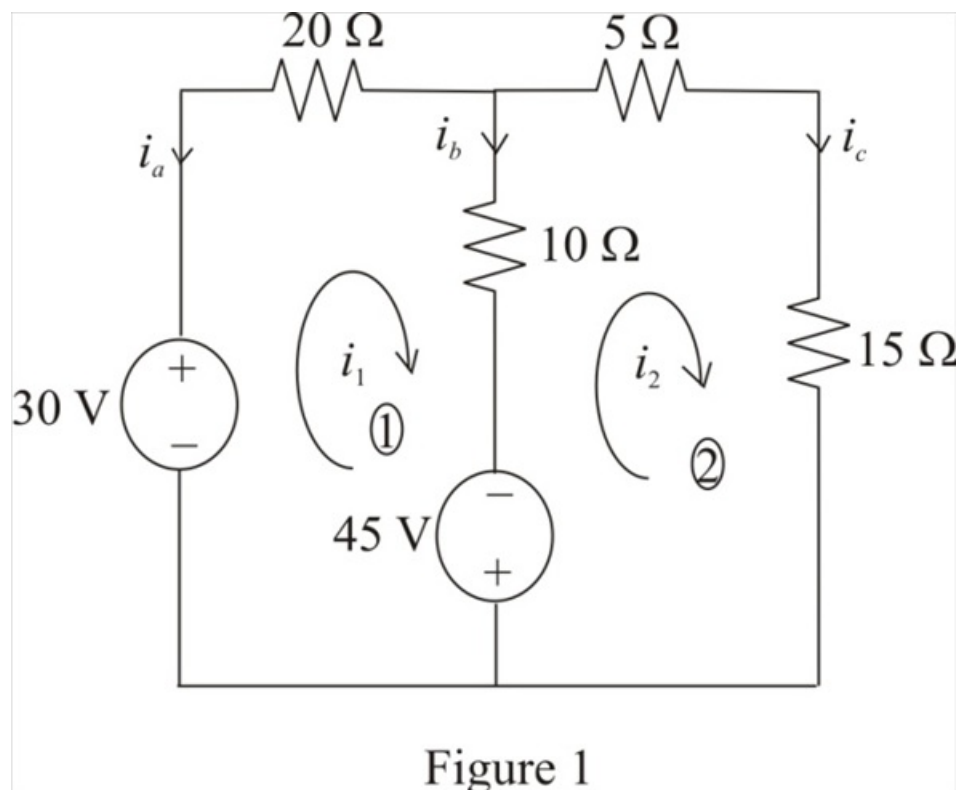


Step-by-step solution

Step 1 of 6

Refer to Figure 3.84 in the text book.

Identify the loops and currents in the circuit.



Step 2 of 6

Apply mesh analysis in loop-1.

Apply Kirchhoff's voltage law, the algebraic sum of all voltages around a closed path (or loop) is zero.

For mesh 1, Kirchhoff's voltage law gives,

$$-30 + i_1(20) + (i_1 - i_2)(10) - 45 = 0$$

$$20i_1 + 10i_1 - 10i_2 - 75 = 0$$

$$30i_1 - 10i_2 - 75 = 0$$

$$30i_1 = 10i_2 + 75$$

$$i_1 = \frac{10i_2 + 75}{30} \dots\dots (1)$$

Therefore, the expression for the current i_1 is, $\frac{10i_2 + 75}{30}$.

Step 3 of 6

Apply mesh analysis in loop-2.

For mesh 2, Kirchhoff's voltage law gives,

$$45 + i_2(5) + i_2(15) + (i_2 - i_1)(10) = 0$$

$$45 + 5i_2 + 15i_2 + 10i_2 - 10i_1 = 0$$

$$45 + 30i_2 - 10i_1 = 0$$

Substitute $\frac{10i_2 + 75}{30}$ for i_1 .

$$45 + 30i_2 - 10\left(\frac{10i_2 + 75}{30}\right) = 0$$

$$45 + 30i_2 - \frac{10i_2 - 75}{3} = 0$$

$$\frac{135 + 90i_2 - 10i_2 - 75}{3} = 0$$

$$135 + 90i_2 - 10i_2 - 75 = 0$$

$$80i_2 + 60 = 0$$

$$80i_2 = -60$$

$$i_2 = \frac{-60}{80}$$

$$i_2 = -0.75 \text{ A}$$

Therefore, the value of the current i_2 is, -0.75 A .

The value of the i_e is $\boxed{-0.75 \text{ A}}$ since, the value of i_e is equal to i_2 .

Step 4 of 6

Recall equation (1).

$$i_1 = \frac{10i_2 + 75}{30}$$

Substitute -0.75 A for i_2 .

$$\begin{aligned} i_1 &= \frac{10(-0.75) + 75}{30} \\ &= \frac{-7.5 + 75}{30} \\ &= \frac{67.5}{30} \\ &= 2.25 \text{ A} \end{aligned}$$

Therefore, the value of the current i_1 is, 2.25 A .

Step 5 of 6

Calculate the value of the branch current i_a .

$$i_a = -i_1$$

Substitute 2.25 A for i_1 .

$$i_a = -2.25 \text{ A}$$

Therefore, the value of the i_a is, $\boxed{-2.25 \text{ A}}$.

Step 6 of 6

Calculate the value of the branch current i_b .

$$i_b = i_1 - i_2$$

Substitute 2.25 A for i_1 and -0.75 A for i_2 .

$$\begin{aligned} i_b &= 2.25 - (-0.75) \\ &= 2.25 + 0.75 \\ &= 3 \text{ A} \end{aligned}$$

Therefore, the value of the branch current i_b is, $\boxed{3 \text{ A}}$.